23-202-0613 OPTIMIZATION TECHNIQUES

Course Outcomes:

On completion of this course the student will be able to:

1. *To understand the need and origin of the optimization methods.*
2. *To get a broad picture of the various applications of optimization methods used in engineering.*
3. *To define optimization problem and its various components.*

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	2	3	2	2	2						1	2	2	1	1	1
CO2	3	3	3	2	2							2	2	2	1	1
CO3	3	3	2	2	2							2	3	1	1	1

1-Slightly;2-Moderately;3-Substantially

Module I

Introduction: Engineering applications of optimization, Formulation of design problems as mathematical programming problems, objective function, constraints, classification of optimization problems/techniques.

Optimization techniques: Classical optimization, unconstrained single and multivariable minimization- necessary and sufficient conditions for optimality, uni-modality, convexity.

Module II

Linear programming problems-I: Mathematical formulation of LP Problems, slack, surplus and artificial variables. Reduction of a LPP to the standard form, feasible solutions. Graphical solution method, simplex algorithm and solution using tabular method, optimality conditions and degeneracy. Duality in linear programming

Module III

Transportation Problem: Formulation of transportation problem, Basic feasible solution using different methods- East West corner method, Vogel approximation method, Optimality methods, MODI method, Unbalanced transportation problem

Game Theory: Introduction, 2- person zero – sum game; Saddle point; Mini-Max and Maxi-Min Theorems (statement only); Graphical solution (2x n, m x 2 game), dominance property. Network path Models: Tree Networks – Minimal Spanning Tree - Prim's Algorithm. Shortest path problems- solution methods – Dijkstra's Method.

Module IV

Nonlinear unconstrained optimization: Single variable optimization methods- Fibonacci search method, Newton- Raphson method.

Multi-variable methods- Hook-Jeeves pattern search method, Cauchy's (steepest descent) method.

Modern methods of optimization: Introduction to Genetic algorithm, Cross over, Mutation, Reproduction, Simple examples of applications in electronics engineering

Text Books:

1. H.A. Taha, "Operations Research", 5/e, Macmillan Publishing Company, 1992.
2. Kalynamoy Deb. "Optimization for Engineering Design- Algorithms and Examples",

Prentice-Hall of India Pvt. Ltd., New Delhi
3. Singiresu S Rao, “Engineering optimization Theory and Practice”, New Age International, 2009

References:

1. A. Ravindran, D. T. Phillips, J. J. Solberg, Operations Research – Principles and Practice, John Wiley and Sons.
2. Ashok D Belegundu, Tirupathi R Chandrupatla, “Optimization concepts and Application in Engineering”, Pearson Education.
3. Hadley, G. “Linear programming”, Narosa Publishing House, New Delhi
4. J. S. Arora, Introduction to Optimum Design, McGraw-Hill Book Company.
5. Kanti Swarup, P.K.Gupta and Man Mohan, Operations Research, Sultan Chand and Sons
6. Papalambros & Wilde, Principles of Optimal Design, Cambridge University Press, 2008